The Delta Method

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Introduction

The delta method is how standard errors propagate through smooth transformations. Given an asymptotically normal estimator $\hat{\theta}$ of θ, the delta method gives the asymptotic distribution of $g(\hat{\theta})$ for a differentiable g. It is how a SE for the log odds ratio becomes a SE for the odds ratio, how a SE for a coefficient becomes a SE for a predicted probability, and how a relative risk acquires a confidence interval from an SE on the log scale.

Prerequisites

The reader should understand asymptotic normality and first-order Taylor expansions.

Theory

Suppose $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} \mathcal{N}(0,\sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$. The delta method states

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2 [g'(\theta)]^2\right).$$

Equivalently, $g(\hat{\theta})$ is approximately $\mathcal{N}(g(\theta), [g'(\theta)]^2\sigma^2/n)$ for large n. The estimated standard error of $g(\hat{\theta})$ is thus

$$\widehat{\mathrm{SE}}[g(\hat{\theta})] = |g'(\hat{\theta})| \cdot \widehat{\mathrm{SE}}(\hat{\theta}).$$

The multivariate version: if $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}(0,\Sigma)$ and $g : \mathbb{R}^p \to \mathbb{R}$ is differentiable with gradient $\nabla g(\theta)$, then

$$\sqrt{n}(g(\hat{\boldsymbol{\theta}}) - g(\boldsymbol{\theta})) \xrightarrow{d} \mathcal{N}\left(0, \nabla g(\boldsymbol{\theta})^\top \Sigma \nabla g(\boldsymbol{\theta})\right).$$

For functions mapping $\mathbb{R}^p \to \mathbb{R}^q$, the Jacobian $J_g(\theta)$ replaces the gradient, and the asymptotic variance is $J_g \Sigma J_g^\top$.

Common applications.

- log odds ratio to odds ratio: if $\widehat{\log \mathrm{OR}}$ has SE s, then $\widehat{\mathrm{OR}} = e^{\widehat{\log \mathrm{OR}}}$ has SE $s \cdot \widehat{\mathrm{OR}}$ (approximately).
- rate ratio from regression coefficients on the log scale.
- relative risk from predicted probabilities in a logistic regression.
- coefficient of variation σ/μ from $\hat{\sigma}$ and $\hat{\mu}$.

Assumptions

- g differentiable at θ with non-zero derivative (first-order delta method). If $g'(\theta) = 0$, use the second-order delta method, which gives a chi-squared asymptotic distribution.
- $\hat{\theta}$ is asymptotically normal. Without this, the delta method does not apply.

R Implementation

```
library(msm)

set.seed(2026)
n <- 200
x <- rnorm(n, mean = 5, sd = 2)
theta_hat <- mean(x)
se_theta  <- sd(x) / sqrt(n)

g  <- function(t) t^2
gp <- function(t) 2 * t

g_hat <- g(theta_hat)
se_g  <- abs(gp(theta_hat)) * se_theta
c(estimate = g_hat, se = se_g)

deltamethod(~ x1^2, mean = theta_hat, cov = se_theta^2)
```

The `msm::deltamethod()` function computes the delta-method SE symbolically from a formula.

Output & Results

```
estimate       se
 24.84     1.408
```

```
[1] 1.408
```

The two approaches agree: the SE of $\hat{\theta}^2$ is about 1.41, computed either by hand or by `deltamethod()`.

Interpretation

In practice, most regression outputs report coefficients on a scale where an SE is available (log odds, log rate), and researchers need effects on the original scale. Delta method is the standard tool for this translation. A 95% CI for the odds ratio is usually constructed as $\exp(\hat{\beta} \pm 1.96\widehat{\mathrm{SE}}(\hat{\beta}))$, which is the delta method on the log scale combined with back-transformation – often more accurate than applying delta method directly to the OR.

Practical Tips

- When the transformation is non-linear (exp, inverse, square), back-transformation of a symmetric CI on the transformed scale is usually preferred over delta-method CIs on the original scale.
- Check that $g'(\theta) \neq 0$; if it is zero, second-order delta method is needed, and the asymptotic distribution becomes chi-squared.
- `msm::deltamethod()` and `car::deltaMethod()` both implement symbolic delta method and are convenient when the function is complicated.
- The delta method gives only the asymptotic variance; for small samples, it is an approximation and may undercover.
- For multivariate transformations (multi-output), use the Jacobian form; scalar delta-method formulas can be seriously misleading in multivariate settings.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**

- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr